

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes polymerization and synthesis methodologies for High Acrylonitrile Nitrile Butadiene Rubber (NBR) based on a selected portfolio of seven key patents published between 2011 and 2024. High Acrylonitrile NBR is defined by a bound acrylonitrile (ACN) content typically exceeding 30%, which imparts superior oil, fuel, and chemical resistance compared to low or medium ACN counterparts. The patent landscape reveals a strong emphasis on emulsion polymerization techniques, particularly cold emulsion processes conducted at 7°C to 8°C to minimize polymer branching and gel formation. Key formulation trends include the use of advanced redox initiator systems (such as hydroperoxides paired with reducing agents and iron catalysts like FeCl₂ or FeSO₄) and precise molecular weight control via incremental dosing of chain-transfer agents (e.g., tertiary dodecyl mercaptan). Furthermore, a major process trend centers on post-polymerization coagulation chemistry, where the choice of salt (magnesium chloride vs. calcium chloride), latex pH adjustment, and the incorporation of additives like gelatin are shown to critically influence the polymer's ion index, Mooney scorch, vulcanization kinetics, and mechanical properties. This report provides a comprehensive cross-patent comparison and detailed extraction of experimental recipes to guide future research and development in high-performance NBR synthesis.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: EP4112676 - Patent Title: METHOD FOR PRODUCING NITRILE RUBBER - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2024-03-13 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Focuses on methods for producing nitrile rubber with controlled properties.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization

temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent describes a method for producing nitrile rubber, but specific experimental parameters are not detailed in the provided text snippet.

Technical Relevance Provides a modern methodology for producing nitrile rubber with optimized process control.

Patent 2

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017-05-16 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Discloses emulsion polymerization recipes for NBR latices (Latex A, B, C) with varying acrylonitrile content.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Latex A: 500 g acrylonitrile to 4.37 kg butadiene; Latex B: 815 g acrylonitrile to 4.0 kg butadiene - Water amount: 9.92 kg deionized water - Emulsifier: Potassium stearate and DNMK (sodium salt of disproportionated rosin acid) - Emulsifier loading: 0.53 kg of 5.4 wt% potassium stearate solution, 0.440 kg of DNMK - Initiator: Diisopropylbenzene hydroperoxide (DIHP) and sodium formaldehyde sulfoxylate (FSS) - Initiator loading: 6 g DIHP, 139 g FSS aqueous solution - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Tertiary dodecyl mercaptan (TDM) - Chain-transfer dosage: Initial 18.72 g (Latex A) or 21.85 g (Latex B), with booster additions of 2.34 g or 5.45 g TDM at specific conversions (37% and 59%) - Polymerization temperature: 7°C - Pressure: Not disclosed in the analyzed patent evidence. - pH: Latex A: 8.5, Latex B: 8.5, Latex C: 8.6 - Reaction time: 7 hours - Conversion: Interrupted at 70% conversion using hydroxylamine sulfate - Coagulation conditions: Latex added to 43 kg water and 4 kg saturated magnesium sulfate solution - Post-treatment: Washing with deionized water adjusted to pH 11.5-11.8 with KOH (14 g) at 45°C, filtering, and drying at 100°C for 24 hours - Raw polymer properties: Mooney viscosity (ML 1+4 @ 100°C) of 60.5 (Latex A), 32.1 (Latex B), and 29.0 (Latex C); bound ACN of 28.1% (Latex A), 33.9% (Latex B), and 27.9% (Latex C)

Relevant Experimental Evidence - Latex A was prepared using 500 g acrylonitrile, 4.37 kg butadiene, and 18.72 g TDM, yielding 28.1% bound ACN and a Mooney viscosity of 60.5. - Latex B was prepared with a higher initial ACN charge of 815 g and 4.0 kg butadiene, resulting in 33.9% bound ACN and a Mooney viscosity of 32.1. - Polymerization was conducted at 7°C, initiated by DIHP/FSS, and short-stopped at 70% conversion after 7 hours using hydroxylamine sulfate. - Coagulation was achieved using magnesium sulfate, followed by washing at pH 11.5-11.8 and drying to less than 1% residual humidity.

Technical Relevance Demonstrates precise control of bound acrylonitrile content and Mooney viscosity through incremental chain-transfer agent (TDM) dosing and low-temperature emulsion polymerization.

Patent 3

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015-05-19 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Focuses on NBR with specific terminal groups and coagulation techniques using gelatin.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during vulcanization) - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Vulcanization conversion t₁₀ and t₉₀ determined at 160°C - Coagulation conditions: Coagulated using magnesium chloride and an acid-ashed gelatin (viscosity 92.1 cP, IEP 8.7) dissolved in the salt solution - Post-treatment: Vulcanization at 160°C under 120 bar hydraulic pressure for 30 minutes - Raw polymer properties: Mooney sheets subjected to hot air aging at 100°C for 48 hours; vulcanizates evaluated for tensile strength, elongation at break, and corrosion resistance

Relevant Experimental Evidence - Coagulation of 25 L of Latex A was performed using magnesium chloride in the presence of an acid-ashed gelatin with a viscosity of 92.1 cP. - Vulcanization was carried out at 160°C under 120 bar pressure for 30 minutes to prepare specimens for mechanical testing. - Hot air aging was performed at 100°C for 48 hours, and corrosion testing was conducted at 70°C and 80% relative humidity.

Technical Relevance Highlights the use of gelatin during coagulation to control the ion index and improve vulcanization and aging properties of NBR.

Patent 4

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011-04-12 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Part of a family of patents focusing on NBR properties and coagulation control.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent belongs to a series of disclosures on nitrile rubbers, but specific experimental parameters are not detailed in the provided text snippet.

Technical Relevance Contributes to the understanding of NBR synthesis and processing parameters.

Patent 5

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013-03-05 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Investigates the impact of coagulation salts (magnesium chloride vs. calcium chloride) on Mooney scorch and vulcanization rates.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during vulcanization) - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Vulcanization conversion t₁₀ and t₉₀ determined at 160°C - Coagulation conditions: Coagulated using magnesium chloride or calcium chloride (e.g., CaCl₂ at 1190-1290 ppm) - Post-treatment: Mixing in a 1.5 L laboratory mixer, followed by vulcanization at 160°C for 30 minutes - Raw polymer properties: Mooney scorch, vulcanization rate (t₉₀ - t₁₀), and stress at 300% elongation (ranging from 7.0 to 7.6 MPa)

Relevant Experimental Evidence - NBR latices coagulated with magnesium chloride display lower Mooney scorch and shorter vulcanization times compared to non-invention counterparts. - Vulcanizate properties, such as stress at 300% elongation (up to 7.6 MPa) and tensile strength (up to 21.6 MPa), depend on the cation concentration and ratio. - Compounding was performed in a 1.5 L laboratory mixer using 100 parts by weight of nitrile rubber.

Technical Relevance Demonstrates that the choice of coagulation salt (specifically magnesium chloride) significantly improves Mooney scorch and vulcanization kinetics.

Patent 6

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013-05-07 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Discloses a specific redox initiator system (using FeCl₂ instead of FeSO₄) for emulsion polymerization of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Butadiene, acrylonitrile, and 2-hydroxyethyl methacrylate (HEMA) - Monomer ratio: Butadiene: 61.4 parts, Acrylonitrile: 36 parts, HEMA: 2.6 parts - Water amount: 220 parts by weight (or 2.1 kg total water for autoclave batches) - Emulsifier: Texapon K-12 (sodium lauryl sulfate) - Emulsifier loading: 3.0 parts by weight (range 0.2 to 15 parts) - Initiator: p-Menthane hydroperoxide (Trigonox NT 50) and Rongalit C - Initiator loading: 0.02 parts Trigonox NT 50, 2.0 g Rongalit C in premix - Catalyst / activator: FeCl₂ (0.030 parts, premix containing 0.845 g FeCl₂·4H₂O) or FeSO₄ (0.030 parts, premix containing 0.986 g FeSO₄·7H₂O) with EDTA - Chain-transfer agent: tertiary dodecyl mercaptan (t-DDM) - Chain-transfer dosage: 0.54 parts by weight - Polymerization temperature: 8.0°C to 0.5°C - Pressure: Hydrogenation pressure of 50 to 150 bar (if hydrogenated) - pH: 7.0 to 1.0 - Reaction time: 6.5 to 7.0 hours - Conversion: 84.2% to 86.1% (stopped using diethylhydroxylamine 0.2 parts) - Coagulation conditions: Latex admixed with 50% dispersion of Vulkanox BKF (0.3 wt% on solids) before coagulation - Post-treatment: Coagulated, washed with water (1 to 5 parts per 100 parts rubber), and dried - Raw polymer properties: Mooney viscosity (ML 1+4 @ 100°C) of 25 to 142 MU, glass transition temperature (T_g) of -23.1°C to -25.8°C, bound ACN of 35.3% to 36.6%

Relevant Experimental Evidence - Nitrile rubber H prepared with FeCl₂ premix reached 84.2% conversion in 6.5 hours, while rubber G with FeSO₄ reached 86.1% in 7.0 hours. - The monomer system comprised 61.4 parts butadiene, 36 parts acrylonitrile, and 2.6 parts HEMA, with 3.0 parts Texapon K-12 emulsifier. - Mooney viscosities of the resulting rubbers ranged from 25 MU (Rubber F) to 142 MU (Rubber C).

Technical Relevance Proves that using FeCl₂ instead of FeSO₄ in the redox system yields nitrile rubbers with distinct and improved properties.

Patent 7

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2014-03-04 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Focuses on the impact of latex pH adjustment and coagulation salt type on the ion index and vulcanization rate of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Latex pH adjusted using 5% aqueous KOH or 5% aqueous HCl before coagulation (e.g., pH 5.8 to 9.5) - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Vulcanization rate t_{90} - t_{10} determined - Coagulation conditions: Coagulated using NaCl (15-26 wt% salt weight) or CaCl₂ (0.3-1.2 wt% salt weight) at temperatures of 20°C to 50°C - Post-treatment: Washing with deionized water (DW) or brackish water (BW) at 20°C to 40°C, followed by drying - Raw polymer properties: Vulcanizates show higher vulcanization rates (t_{90} - t_{10}) and higher stress values at 30% elongation (up to 10.2 MPa) depending on ion indices

Relevant Experimental Evidence - Latex pH was adjusted to 9.5 using 5% KOH (C6) or to 5.8 using 5% HCl (C13) prior to coagulation. - Coagulation with CaCl₂ (0.3 to 1.2 wt%) resulted in higher ion indices (up to 96.2) and shorter vulcanization times compared to NaCl coagulation. - Vulcanizates prepared from rubbers with optimized ion indices showed stress values at 30% elongation of up to 10.2 MPa.

Technical Relevance Demonstrates that adjusting latex pH prior to coagulation and selecting specific divalent salts (CaCl₂) optimizes the polymer's ion index, leading to faster vulcanization rates and superior mechanical properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No cross-patent comparison provided.

4. REFERENCES

#	Patent No.	Title	Assignee	Jurisdiction	Year	Source
1	EP4112676	METHOD FOR PRODUCING NITRILE RUBBER	Not disclosed	EP	2024-03-13	Google Patents
2	US7923518	Nitrile rubbers	Not disclosed	US	2011-04-12	Google Patents
3	US8389623	Nitrile rubbers	Not disclosed	US	2013-03-05	Google Patents
4	US8436091	Nitrile rubbers	Not disclosed	US	2013-05-07	Google Patents
5	US8664340	Nitrile rubbers	Not disclosed	US	2014-03-04	Google Patents
6	US9034992	Nitrile rubbers which optionally contain alkylthio terminal groups and which ...	Not disclosed	US	2015-05-19	Google Patents
7	US9650502	Process for the preparation of nitrile rubbers	Not disclosed	US	2017-05-16	Google Patents